

✓ Lab 2 — Exercise: AI Tools & the Data Mining Pipeline

ITCS 3162 — Introduction to Data Mining

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Complete the exercises below. **Some questions ask you to use an AI assistant** — when you do, you must:

1. Paste the prompt you used in the markdown cell labeled "PROMPT".
2. Paste the AI's response (or key parts) in the cell labeled "AI RESPONSE".
3. Write your own evaluation in "YOUR EVALUATION".

This is the appropriate-use pattern: prompt, response, your judgment.

Exercise 1 — Identify appropriate use (15 pts)

For each scenario, decide whether the AI use is **appropriate** or **inappropriate** and explain why in one sentence.

#	Scenario
A	A student doesn't understand what "stratified sampling" means and asks an AI to explain it with an example.
B	A student pastes the full text of an exercise question and submits the AI's answer as their reflection.
C	A student has a <code>KeyError</code> they can't figure out and asks the AI what it means.
D	A student asks the AI to write all their markdown answers in this notebook so they can focus on the code.
E	A student writes their own code, then asks the AI to suggest more idiomatic alternatives, then chooses one and explains it.

YOUR ANSWERS:

- **A:** appropriate
- **B:** inappropriate
- **C:** appropriate
- **D:** inappropriate
- **E:** appropriate

Exercise 2 — Rewrite a weak prompt (35 pts)

Below is a weak prompt. Rewrite it so it's specific, scoped, and verifiable. Imagine you have a DataFrame `df` with a column `price` (numeric) and you want to detect rows whose price is unusually high.

Weak prompt: "find weird stuff"

YOUR REWRITTEN PROMPT: I have a pandas dataframe with a column named "price" which is a numeric type. Use sklearn's linear regression model to detect where the price is unusually high.

✓ Exercise 3 — Prompt, response, evaluate (20 pts)

Use any AI assistant (ChatGPT, Claude, Copilot chat, Gemini, etc.) to answer this question:

"In scikit-learn, what is the difference between `fit`, `transform`, and `fit_transform`? When would I use each?"

Fill in all three cells below.

PROMPT (what you actually typed): In scikit-learn, explain the difference between `fit`, `transform`, and `fit_transform`? When would I use each?

AI RESPONSE (paste it here, or summarize the key points):

These three methods are the backbone of data preprocessing in scikit-learn (used in classes like StandardScaler, PCA, or SimpleImputer). To understand them, it helps to look at a concrete example: standardizing data. To scale a dataset, the algorithm needs to calculate the mean (μ) and standard deviation (σ), and then apply the formula $z = \frac{x - \mu}{\sigma}$ to every data point. Here is how the three methods break that down:

- Method** **What it does** **What it returns**
- fit(X)** Learns the parameters. It calculates the underlying statistics (like the mean and standard deviation of X) but does not change the data. The transformer object itself.
- transform(X)** Applies the math. It uses the parameters previously learned by fit to modify X. A new, transformed array.
- fit_transform(X)** Does both. It learns the parameters from X and immediately modifies X. Under the hood, this is often faster than calling them separately. A new, transformed array.

It also gave me a couple golden rules where it talked about where and how to use training, validation and testing data

YOUR EVALUATION — answer all three:

1. Was the response correct? How did you verify?

- The response was correct, I went over the documentation of scikit-learn of the diffrenet functions, specifically the fit transform and fit_transform functions. The ai got a generally good idea of what those functions do.
2. Was anything missing or unclear?
- I used Gemini Pro and it gave a decently clear and correct result for me.
3. In your own words (no copy-paste), explain the difference between `fit` and `transform`.
- Fit is used to learn information or the statistics about the dataset. Transofrm is the functon that actually applies what we learned to actually change the dataset, to become more usefull.

✓ Exercise 4 — Map the pipeline (15 pts)

Below is a list of activities. For each, write which **pipeline stage** it belongs to (Problem understanding / Load / Explore / Preprocess / Model / Evaluate).

#	Activity	Stage
A	Running <code>df.describe()</code> to see column means and quartiles	Explore
B	Choosing whether to use a decision tree or logistic regression	Problem Understanding
C	Replacing missing ages with the median age	Preprocess
D	Talking to a stakeholder to clarify what "customer churn" means	Problem Understanding
E	Computing accuracy and confusion matrix on the held-out test set	Model
F	Reading a CSV with <code>pd.read_csv("data.csv")</code>	Load
G	One-hot encoding a <code>country</code> column	Preprocess
H	Plotting a histogram of <code>salary</code>	Evaluate

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# Exercise 5 – Build your own mini-pipeline (20 pts)
#
# Using the wine dataset (built into sklearn), build a complete pipeline
# 1. Loads the data as a DataFrame
# 2. Prints the shape and the class distribution of the target
# 3. Splits into 70/30 train/test with random_state=42, stratified on
# 4. Builds a sklearn Pipeline with StandardScaler + LogisticRegression
# 5. Fits on the training data and reports test accuracy
#
# Fill in the TODOs below.

import pandas as pd
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
```

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from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

# 1. LOAD
wine = load_wine(as_frame=True)
df = wine.frame
print(df.shape)
print(df["target"].value_counts())

# 2. SPLIT
X = df.drop(columns="target")
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.30)

# 3. PIPELINE
# TODO: build the Pipeline with two steps named "scale" and "model"
pipe = Pipeline([("scale", StandardScaler()), ("model", LogisticRegression)])

# 4. FIT + SCORE
# TODO: fit on the training data
pipe.fit(X_train, y_train)

# TODO: print the test accuracy with 3 decimal places
accuracy = pipe.score(X_test, y_test)
print(accuracy)

```

```

(178, 14)
target
1      71
0      59
2      48
Name: count, dtype: int64
0.9814814814814815

```

Exercise 6 — Reflection (15 pts)

In 4–6 sentences, answer **both**:

1. What's one task in this course you think AI tools would help you with the most, and one task where you'd intentionally not use them? Why?
2. Looking at the six-stage pipeline, which stage do you expect to find hardest and why?

YOUR ANSWER:

1. One task I think I could use AI is to help me with syntax of some of the functions I'll write, mainly as a time saver, I could ask it for templates of a specific preprocessing

pipeline for example. One task I would intetinally not use AI is when it comes to data preprocessing for example, I would want to truly understand what the parameters or approches I pick, and how they effect the data.

2. Preprocessing seems like the hardest part of the pipeline to me, minly becuae it is not often obvious what part of the data can effect the overall final result of the prediction or visulization. Choosing the corect data to drop or replace is the more challenging part.

Submission checklist

- ☐ Name and date filled in at the top
- ☐ All `YOUR ANSWER` / `TODO` prompts completed
- ☐ Exercise 3 includes a PROMPT, AI RESPONSE, and YOUR EVALUATION
- ☐ Code cell in Exercise 5 runs and prints test accuracy
- ☐ **Restart & Run All** completes without errors
- ☐ Downloaded as `.ipynb` and submitted via Canvas